

Multiple Choice

1. **Answer:** D

Options: A) 4.0×10^8 eV and 7.7×10^6 eV; B) 4.2×10^8 eV and 8.1×10^6 eV; C) 4.6×10^8 eV and 8.9×10^7 eV; D) 4.6×10^8 eV and 8.9×10^6 eV; E) 5.0×10^8 eV and 9.6×10^6 eV

Note: Correct option: D - 4.6×10^8 eV and 8.9×10^6 eV

2. **Answer:** A

Options: A) 30 mi per hr, 60 ft per sec; B) 90 mi per hr, 132 ft per sec; C) 75 mi per hr, 110 ft per sec; D) 60 mi per hr, 55 ft per sec; E) 45 mi per hr, 88 ft per sec

Note: Correct option: A - 30 mi per hr, 60 ft per sec

3. **Answer:** E

Options: A) 150 cm/sec and 7.5 m; B) 350 cm/sec and 17.5 m; C) 450 cm/sec and 22.5 m; D) 250 cm/sec and 12.5 m; E) 200 cm/sec and 10 m

Note: Correct option: E - 200 cm/sec and 10 m

4. **Answer:** A

Options: A) 4.23:1; B) 10.24:1; C) 8.67:1; D) 2.98:1; E) 5.67:1

Note: Correct option: A - 4.23:1

5. **Answer:** A

Options: A) We can't see magnetic fields; B) The corona is blocked by the Earth's atmosphere; C) The corona is hidden by the Sun's photosphere; D) The corona is made up mostly of charged particles which aren't luminous.; E) The Sun's corona is actually invisible to the human eye

Note: Correct option: A - We can't see magnetic fields

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'tension.'

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '2100.0'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '910⁷yr'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '2.5 cm'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 12. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 15432.98 Joules. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key